MOHAMMAD SHERIFF MEHMOOD

Senior Data Scientist · NLP & Generative AI Engineer

+91 79999 37693 | mdsheriff2702@gmail.com | LinkedIn | github.com/sheriff786 | Portfolio

Senior Data Scientist with 6+ years delivering production ML and NLP systems in healthcare, finance, and retail. Deep expertise in Generative AI (RAG pipelines, LLMOps), end-to-end MLOps, and scalable predictive modelling. Self-driven researcher who explores problems beyond assigned scope — built and published authentica, an open-source AI image-authenticity detection library on PyPI now used in production with a client, and documented its architecture in a public technical deep-dive. Track record of measurable business impact: $1M+ revenue uplift, 40% churn reduction, 60% faster model iteration. GCP Certified ML Engineer.

TECHNICAL SKILLS

Languages & GenAI: Python (primary), FastAPI, SQL, PyTorch, TensorFlow, Scikit-learn, LangChain, LangGraph, OpenAI API (GPT models), RAG, LLMOps, Fine-tuning, Prompt Engineering, Agentic Systems (multi-agent orchestration, context engineering, tool-use harnessing), NLP, LLMs

MLOps & Cloud: MLflow, Docker, AWS (SageMaker, Lambda, ECS/Fargate, API Gateway, Step Functions, CDK), GCP, CI/CD, GitHub Actions, Model Versioning & Governance

Document AI & CV: Amazon Textract (Queries/Forms/Tables), Amazon Comprehend (Custom Classification), OpenCV, YOLO, OCR Pipelines, Document Forensics

ML & Statistics: XGBoost, Random Forest, Logistic Regression, Deep Learning, CNN, UMAP, K-Means Clustering, Embedding Models, Feature Engineering, A/B Testing, Hypothesis Testing, Probability & Statistics

Databases & Vector Stores: PostgreSQL, DynamoDB, MongoDB, Vector DB (FAISS, Pinecone)

PROFESSIONAL EXPERIENCE

ValueLabs — *Hyderabad, India*

Senior Data Scientist – AI/ML Engineer — Mar 2025 – Present

Embedded with Businessolver's Data Science team as a full-stack ML Engineer; designed and shipped an automated document analysis pipeline leveraging computer vision, VLMs, and AWS-native tooling to process healthcare documents at scale.

- **Document Analysis Pipeline:** Contributed to document extraction within an intelligent document understanding platform on AWS (Comprehend, Textract, S3, ECS), processing tens of thousands of healthcare documents/week across 20+ document classes; helped evolve the system from OCR/heuristic rules to fine-tuned neural models, reaching high-0.8 to 0.9 F1 on classification and 0.7–0.9 field-level extraction accuracy — significantly cutting manual claim validation time.
- **Target-Table Selection ML:** Developed document-level target-table selection using header semantics, layout, and structural signals; applied group-aware, document-ID-based splits to prevent leakage, benchmarking XGBoost against a heuristic baseline on a held-out 200-document, 653-table dental cohort.
- **Grounded GenAI Extraction:** Built evidence-grounded GenAI extraction where model outputs reference Textract Word IDs, validated via schema/ownership/source checks before rehydrating reviewer-visible values — improving carrier-specific extraction from a ~65-70% rule-based baseline to 90%+ on documented carrier cohorts.
- **Annotation & Evaluation Console:** Built an interactive React UI with REST API for data annotation/validation and evaluation dashboards (confusion matrices, threshold curves, field-level failure analysis), enabling data-driven threshold calibration and streamlining stakeholder review cycles.
- **Hybrid Census Analytics Engine:** Re-architected census analytics with hybrid LLM–Pandas processing, reducing LLM calls from 30–40 to 1 and eliminating token-limit bottlenecks through schema mapping and deterministic analytics.
- **Centralized LLM & Cloud Infrastructure:** Built reusable OpenAI client factories and AWS Secrets Manager integration with singleton/cached clients, unified fallback/configuration, secure credential management, and centralized observability.
- **Production-Grade Hybrid RAG Architecture:** Designed a tenant-isolated hybrid RAG system using Qdrant and OpenAI embeddings with semantic + BM25 retrieval, query rewriting, bounded agentic search, source diversification, and citation-grounded responses.

LTIMindtree — *Pune, India*

Senior Data Scientist – NLP Engineer — Jan 2024 – Mar 2025

- **GenAI Chatbot & RAG System:** Architected a production-grade RAG chatbot (Gemini + FAISS Vector DB) serving 15+ team members with 90% query accuracy and sub-2s response time for enterprise knowledge retrieval.
- **LLMOps Pipeline:** Built an end-to-end LLMOps pipeline using OpenAI, LangChain, ChromaDB, and Sentence-Transformers, cutting model iteration time by 60% and reducing deployment overhead by 50%.

Cognizant Technology Solutions — *Chennai, India*

Programmer Analyst to Data Scientist — Dec 2019 – Jan 2024

Joined as Programmer Analyst on core ML systems; promoted to Data Scientist leading cross-functional AI/NLP initiatives.

- **ML Model Development:** Built and deployed supervised ML models, improving prediction accuracy by 20% and saving $200K in operational costs; improved baselines by 5–10% using Neural Networks, ensemble methods, GridSearchCV, and hyperparameter tuning.
- **AI Customer Support Platform:** Led a 7-member cross-functional team to build an NLP-powered support application, reducing customer churn by 40% and generating $300K in additional quarterly revenue through context-aware intelligent responses.
- **Fraud Detection System:** Reduced false-positive fraud alerts by 20% using Logistic Regression and Random Forest with class-weighted resampling; received the company Innovation Award.
- **Data Platform & BI:** Unified 10+ heterogeneous data sources — including databases, APIs, and third-party feeds — into ML-ready pipelines; built Tableau dashboards tracking 20+ KPIs, improving cross-team reporting efficiency by 30%.

RESEARCH PROJECTS

Authentica – AI Image Authenticity Detection Library View on GitHub → | PyPI | Medium Blog

Python · C2PA · CBOR · COSE · NumPy · SciPy · PyPI · FastAPI

- Built and published authentica, an open-source Python library on PyPI for detecting AI-generated images using a three-layer verification stack: C2PA manifest parsing, invisible watermark detection (DCT/DWT/FFT), and image forensics (ELA, noise residual analysis).
- Implemented cryptographic signature verification (COSE_Sign1/RFC 8152) with JUMBF/CBOR binary parsing for C2PA provenance validation, achieving ~95% combined detection accuracy across all three techniques.
- Published a technical deep-dive blog on Medium documenting the full architecture, from binary container parsing to trustscore computation.
- **Production Impact:** Adopted the library into Businessolver's document-processing pipeline at ValueLabs to automatically flag AI-manipulated images in submitted claims documents, adding a new layer of fraud/authenticity screening to the existing extraction workflow.

Accident & Fall Detection System View on GitHub →

Python · YOLO · OpenCV · CNN · CUDA · SMTP

- Built real-time human fall detection system achieving 95% accuracy using YOLO + CUDA toolkit on edge hardware.
- Designed smart camera surveillance for highway accident detection; automated ambulance dispatch via SMTP alerts, cutting emergency response time by 40%.

Abstractive Text Summarisation View on GitHub →

Python · BERT · NLP · Transformers

- Applied BERT and NLP toolkits to summarise live chat context, improving agent efficiency by 35% and reducing customer response times by 25%.
- Gave agents instant access to previous chat history — reducing handle time and improving resolution quality.

Visa Approval Forecasting (US Immigration) View on GitHub →

Python · ML · MongoDB · Docker · AWS · GridSearchCV

- Predicted US visa approval outcomes using production-grade ML models; performed EDA, feature engineering, and hyperparameter tuning via GridSearchCV.
- Connected MongoDB for data persistence; containerised solution with Docker for AWS deployment.

Disease Classification Mobile App View on GitHub →

TensorFlow · CNN · TFLite · FastAPI · GCP · React

- Achieved 92% accuracy detecting Early/Late Blight in potato crops; quantized CNN with TF Lite — reduced model size by 80% and inference latency by 50% for on-device deployment.
- Deployed serverless inference on Google Cloud Functions (25% cost reduction); served real-time predictions via TF Serving + FastAPI to ReactJS and React Native frontends.

End-to-End Topic Modelling & MLOps on AWS View on GitHub →

Python · LDA · Docker · AWS ECS · MLflow · EC2

- Built LDA topic modelling pipeline on AWS ECS with blue/green deployment — cut processing time by 40% and deployment time by 60%.
- Automated full CI/CD (CodeCommit, CodeBuild, CodeDeploy, CodePipeline); reduced manual effort by 70%.
- Used MLflow for model versioning; ensured high availability via EC2 + ECS load balancing.

EDUCATION

M.Tech in Data Science & Engineering — BITS Pilani, India Oct 2021 – Oct 2023
B.E. in Computer Science & Engineering — OIST, India Aug 2015 – Jun 2019

CERTIFICATIONS & AWARDS

- GCP Professional ML Engineer (Certified Feb 2023)
- 1st Prize – Trizetto Hackathon (Patient Data Integration, 95% approval rate)
- IBM Professional Data Science Certification
- Certified MLOps Developer – Dataiku
- OpenAI Whisper Hackathon – Hearing-impaired accessibility app
- Secretary, Toastmasters Club · Speaker, PyData Conference